NASA SPACE APPS CHALLENGE

Where on Earth?

Earth Event Lab

An iceberg investigation from space



Earthrise, 24 December 1968. NASA / Bill Anders.

Where does the city end?

Trace the dark edge with a finger. What would you add to the map to check your guess?

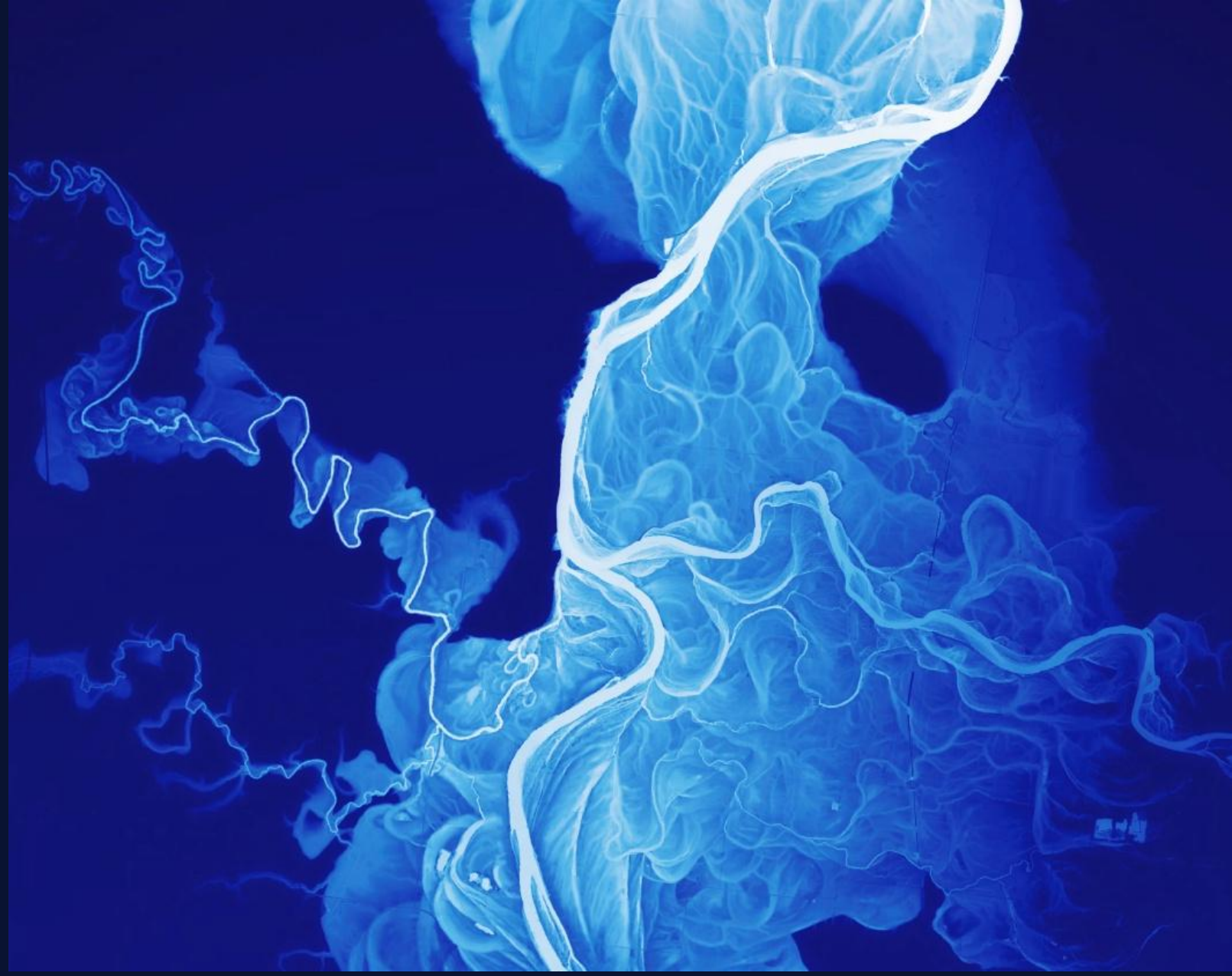


San Francisco Bay and the Central Valley, 25 September 2013. Astronaut photograph, NASA Johnson Space Center.

Can you trace an old river?

Follow a pale loop away from the main channel.

White = lower ground
Dark blue = higher ground



Willamette River, Oregon • Detail of a lidar elevation map by Daniel E. Coe, Oregon Department of Geology and Mineral Industries.

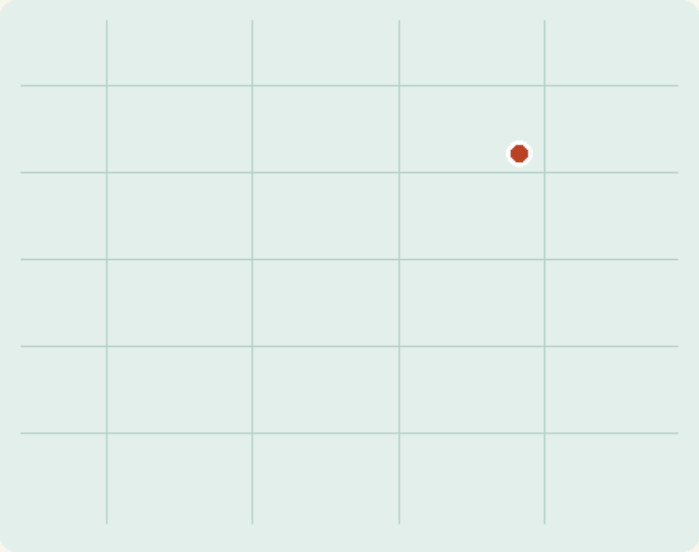


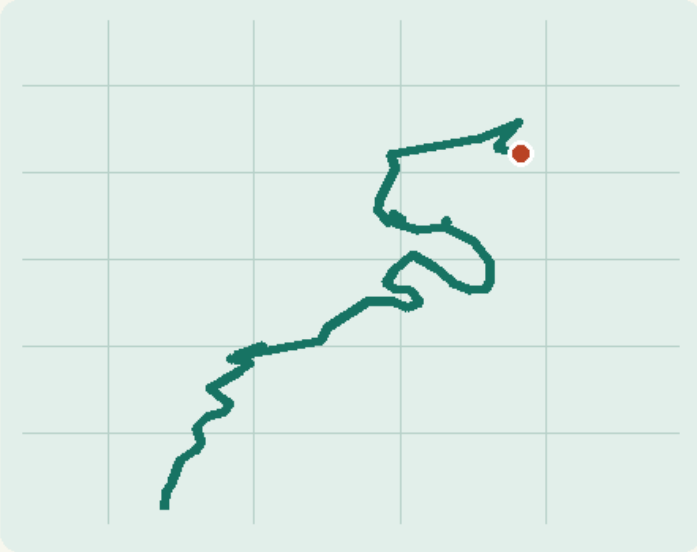
Where does the smoke begin?

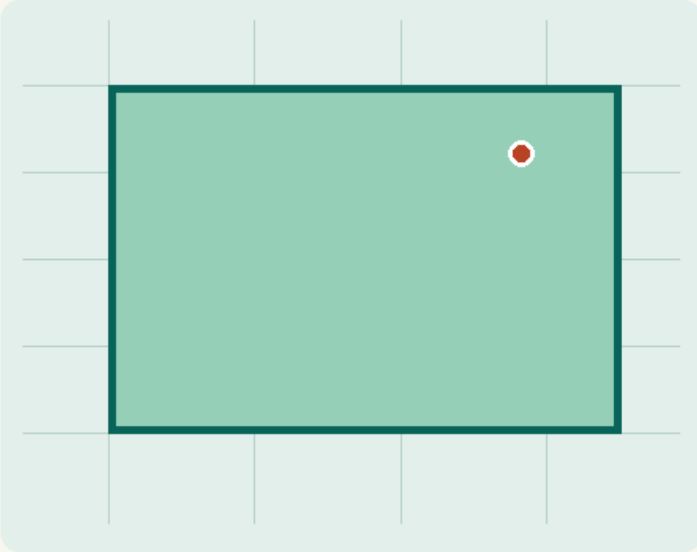
Pick a plume. What would you map to find communities downwind?

Near Novo Progresso, Brazilian Amazon • 19 August 2019 • Terra/MODIS • NASA Earth Observatory, Joshua Stevens.

What shape would you map?







Point

One dated location

Polyline

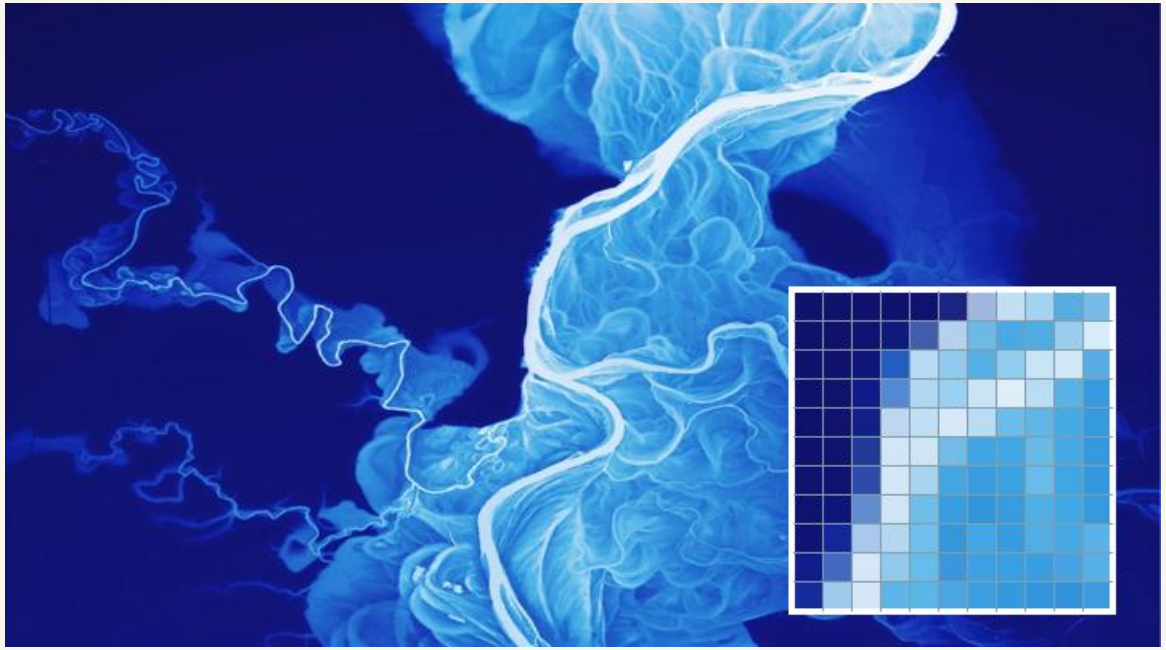
A sequence of positions

Polygon

An area with a boundary

A-23A in the South Atlantic. The green polygon is our study box. Which shape would describe the ice edge?

Pixels or features?



Raster

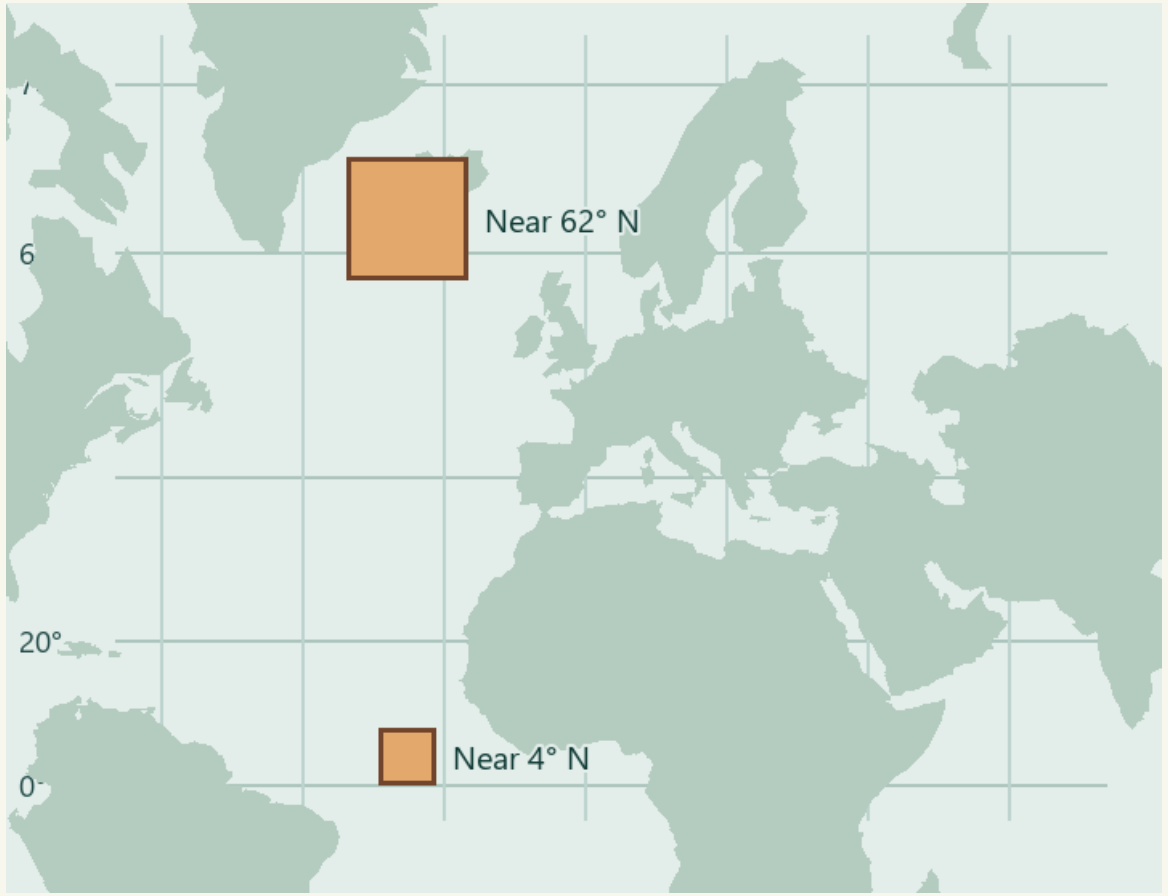
A grid of values. An elevation cell stores height. An image pixel can store brightness.

Vector

Features keep coordinates and attributes. One pin can carry a date and a source.

Which dataset would you use to read ground height? Which would you use to select one named iceberg?

Why does the northern square grow?



Coordinates give a position

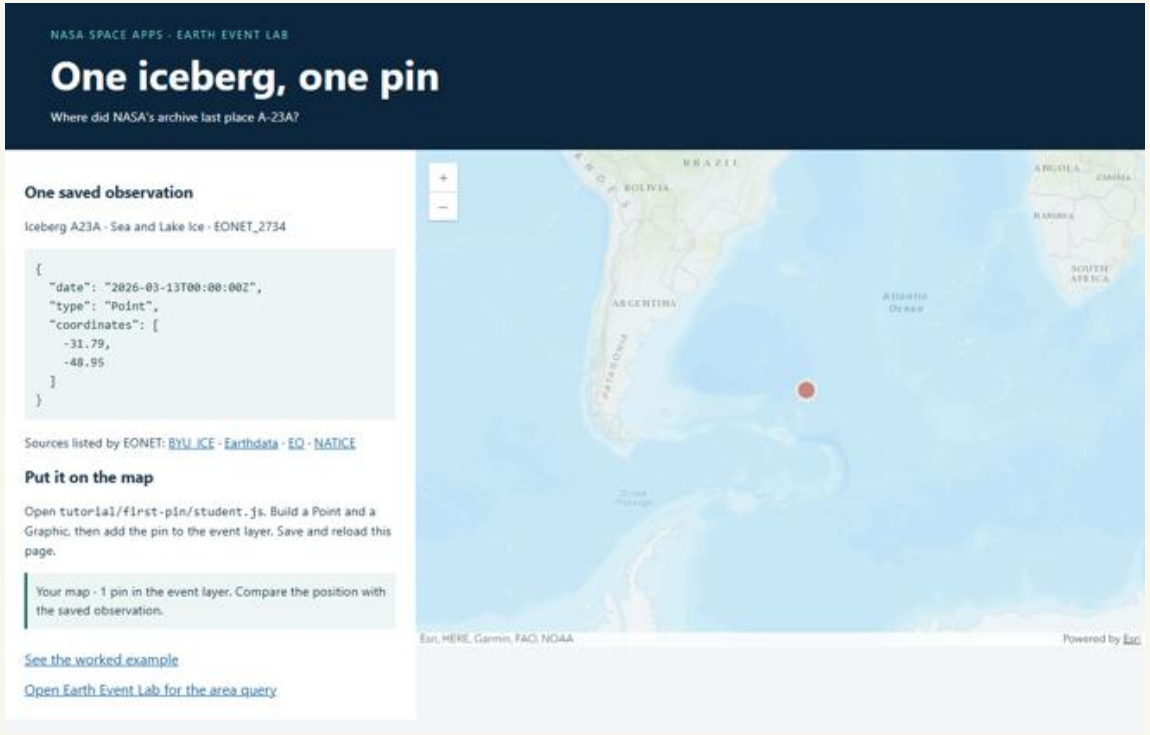
Our WGS 84 values use longitude and latitude in degrees.

A projection draws it flat

Web Mercator enlarges high latitudes. Use an equal-area projection to compare areas.

Both patches are roughly 900 km per side on Earth. Coastlines from Natural Earth.

What loads when you zoom?



Basemap

Land, coastlines and labels give the event a geographic frame.

Tiles

The map requests small pieces at each zoom level. Raster tiles are images. Vector tiles carry features to style.

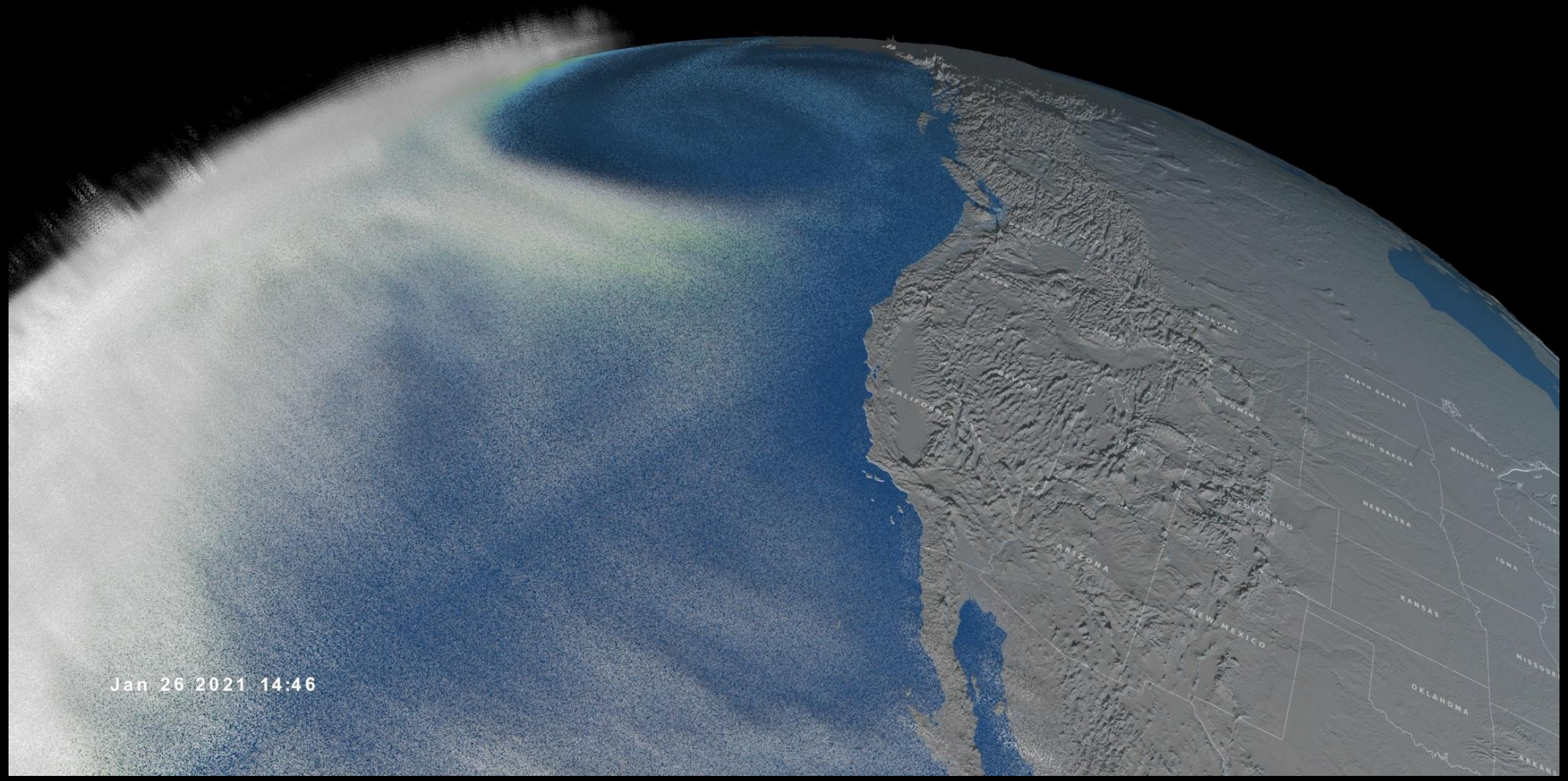
Pan across the ocean. The map requests other tiles. Predict whether the iceberg count changes.

What can a satellite measure from here?

This one studied Earth’s upper atmosphere. What could an instrument detect that our eyes miss?



Upper Atmosphere Research Satellite above Earth • Space Shuttle Discovery, 15 September 1991 • NASA, photo MSFC-9254338.



A river in the sky?

Trace the white band toward land. Which coast would you investigate first?

Pacific coast • 26 January 2021, 14:46 UTC • NOAA-20 atmospheric measurements • NASA's Scientific Visualization Studio.



Why is the ice blue?

Look for a clue at the white rim. Compare guesses with a partner.

International Space Station, 27 December 2025. NASA Johnson Space Center.



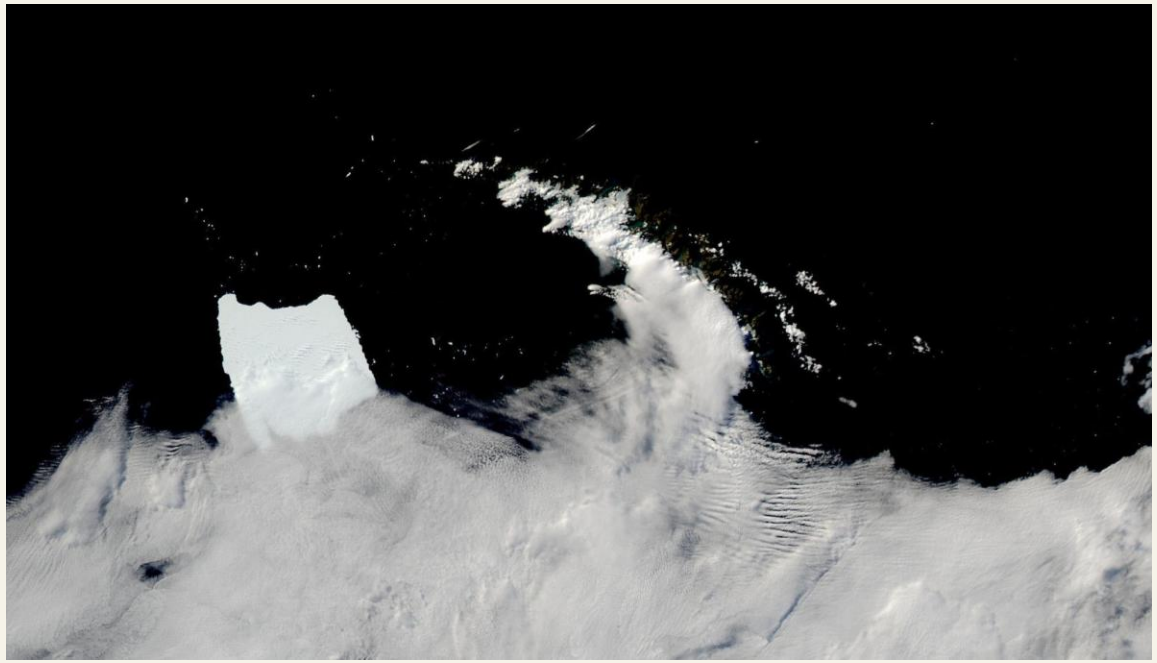
Pools on a floating iceberg

A day earlier, NASA's Terra satellite saw the same iceberg. Meltwater had collected across its surface.

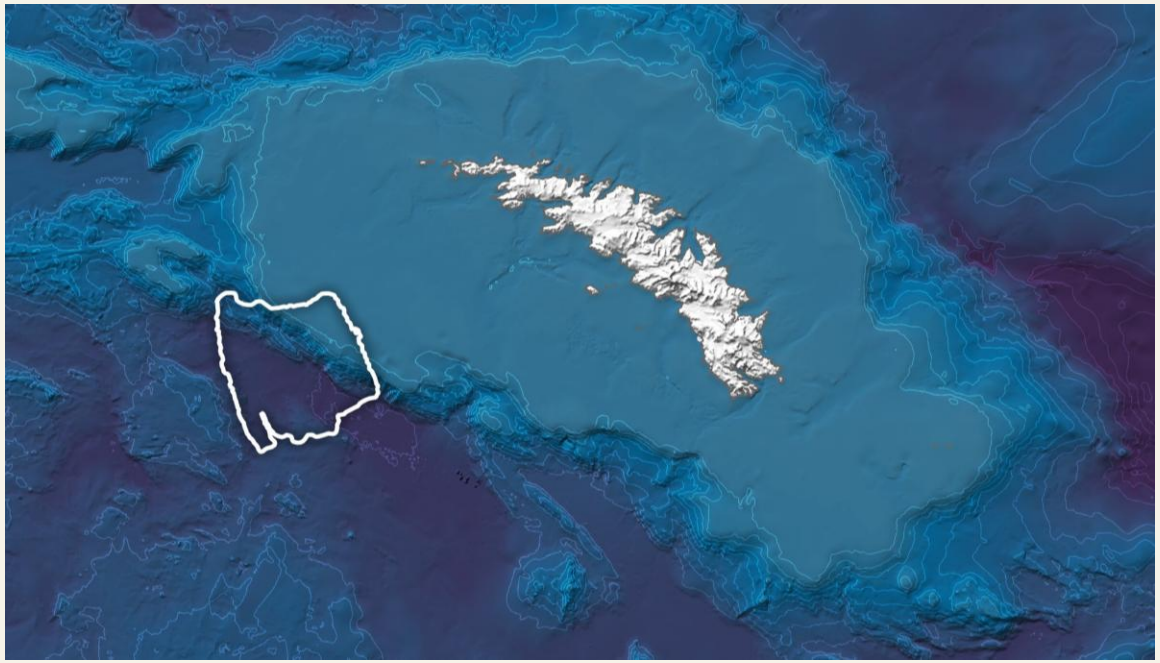
How would you find this piece of ice again once it drifted?

A-23A in the South Atlantic, 26 December 2025. Terra MODIS satellite image, NASA Earth Observatory / Michala Garrison.

Why did the iceberg stop here?



What the satellite saw



What lies under the water

The outline of the ice meets the shallow shelf. A geographic information system (GIS) lets us compare these layers at the same place.

South Georgia, 4 March 2025. Satellite image and seafloor map, NASA Earth Observatory / Wanmei Liang. Aqua MODIS, GEBCO and British Antarctic Survey data.

Who does what on this map?

NASA EONET	Which events were reported, where and when?
ArcGIS Pro	Esri’s desktop GIS for preparing layers and analyzing terrain.
ArcGIS Maps SDK	Esri’s JavaScript tools draw our events in a browser.

Explore EONET reports for wildfires, severe storms, volcanoes or sea and lake ice.

Which mapping tool fits the job?

Esri ArcGIS	Prepare and analyze layers in Pro. Build a web or native GIS app.
Mapbox GL JS	Style an interactive vector map with Mapbox's web renderer.
MapLibre GL JS	Use an open-source renderer with a compatible tile source.
Google Maps Platform	Show a Google map or look up nearby places.

The renderer, the data source and service credentials are separate choices.



Which day does a pin show?

NASA's Earth Observatory Natural Event Tracker (EONET) lists 72 dated positions for A-23A in our sample.

Latest point in the app

13 March 2026

31.79° W, 48.95° S

Track map by Michala Garrison, NASA Earth Observatory, using USNIC and BYU data. The map and the workshop sample cover different observation dates.

Open the workshop files

Workshop repository

Placeholder for the class link

Workshop GitHub

NASA data you can try now

Scan for up to five sea and lake ice records.

Open the NASA feed

Open the workshop folder that contains application, tutorial and sample-data.

The shared file sample-data/eonet-snapshot.json gives everyone the same 48 archived events.

Run the map on your laptop

Open a terminal in the workshop folder.

Windows `py -m http.server 8000`

macOS / Linux `python3 -m http.server 8000`

Open http://localhost:8000/tutorial/first-pin/

Keep the terminal open. You should see Iceberg A23A and a practice map with 0 pins.

Which two numbers place the pin?

Iceberg A23A / EONET_2734 / JSON excerpt

```
{
  "date": "2026-03-13T00:00:00Z",
  "type": "Point",
  "coordinates": [-31.79, -48.95]
}
```

Longitude comes first.
Latitude comes second.

Which hemisphere does each minus sign indicate?

EONET v3 API eonet.gsfc.nasa.gov/api/v3/events

Open the practice map. Find the observation date and one reporting source.

What is GeoJSON?

It is a file format for geographic features. Geometry locates the feature. Properties describe it.

```
{
  "type": "Feature",
  "geometry": { "type": "Point", "coordinates": [-31.79, -48.95] },
  "properties": { "title": "Iceberg A23A", "date": "2026-03-13" }
}
```

Open sample-data/a23a.geojson. Which field would you change to draw a different location?

Build one pin

Open tutorial/first-pin/student.js. The first line is already there.

```
const [longitude, latitude] = observation.coordinates;

const point = new Point({ longitude, latitude });
const pin = new Graphic({ geometry: point, symbol });
eventsLayer.add(pin);
```

Save and reload. Is your marker near 32° W, 49° S?

Which icebergs fall inside our box?

Open Earth Event Lab. Choose Use sample data, then Sea and Lake Ice.

West	East	South	North
−60°	−25°	−65°	−45°

Make a prediction. Enter the four bounds, choose Apply geographic box, and record the names you find.

In this archive, the category contains 12 ice records. Each contributes its latest valid point.

Which iceberg drops out?

Keep the other edges fixed. Move North from −45° to −50°.

Inside the first box	Latest latitude	Observation date
Iceberg A23A	48.95° S	13 Mar 2026
Iceberg A23J	52.86° S	16 Jan 2026
Iceberg D33D	63.78° S	23 Jul 2026

Predict with a partner, then apply the change. Point to the latitude that explains your result.

Latest valid point for each matching event in the 12 November 2026 workshop sample. All three longitudes are inside the original box.

Ask the same question in PostGIS

PostGIS adds geographic types and spatial queries to a PostgreSQL database.

```
SELECT title FROM events
WHERE ST_Intersects(geom,
  ST_MakeEnvelope(-60, -65, -25, -50, 4326));
```

West, south, east, north. 4326 identifies WGS 84.

Our example table has the three ice points from the first box. Which two rows will the query return?

What became of A-23A?

13 March 2026

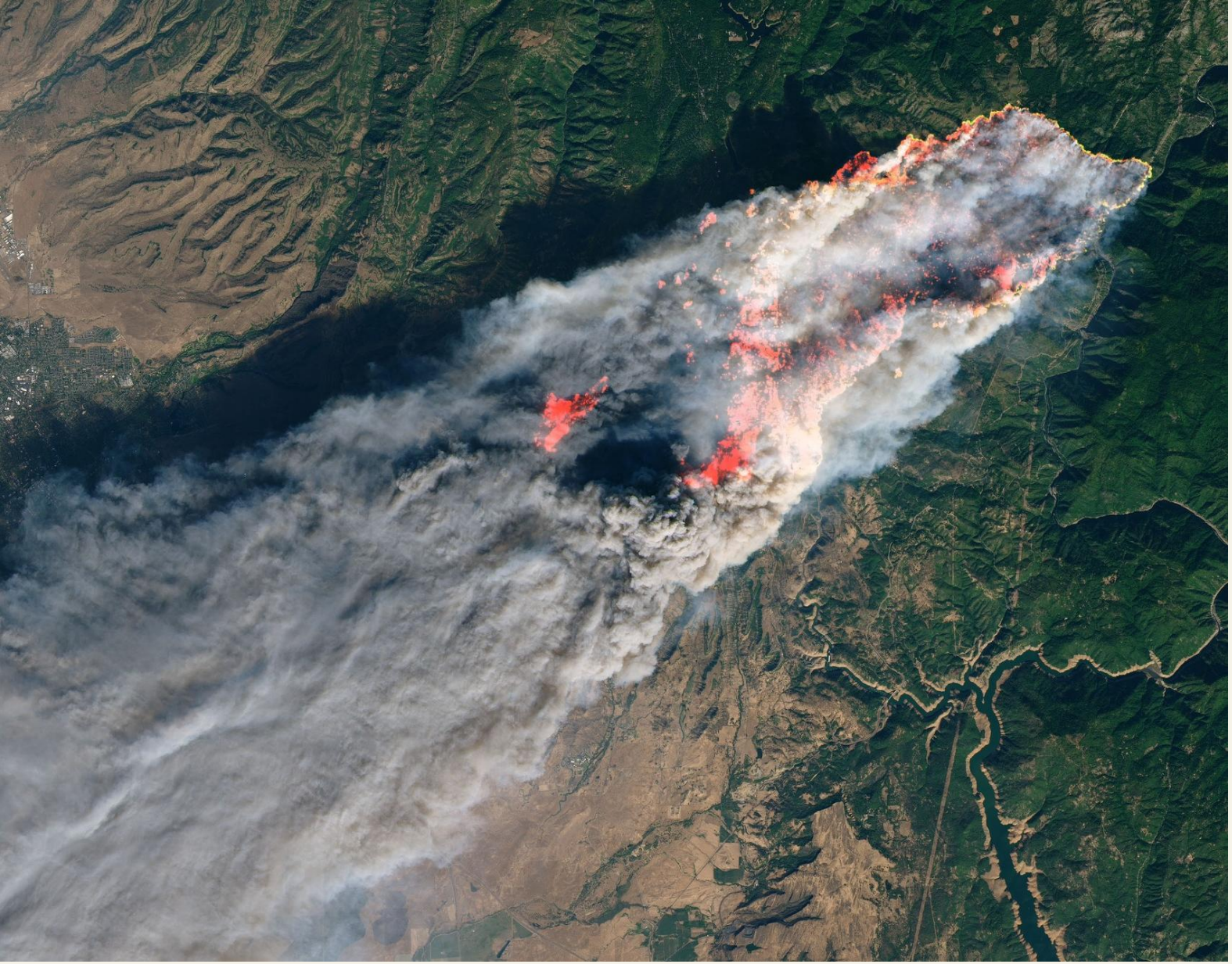
The last position in our sample.

3 April 2026

A satellite saw scattered pieces.



South Atlantic, 3 April 2026. NOAA-21 VIIRS satellite image, NASA Earth Observatory / Michala Garrison.



Which towns were under this smoke?

A fire report gives you a location. What would you layer over this image to investigate a town downwind?

Camp Fire, California, 8 November 2018. Landsat 8 satellite image using visible light and shortwave infrared to highlight fire. NASA Earth Observatory / Joshua Stevens, USGS data.



Where did the ash go next?

An astronaut caught Raikoke erupting. Which observation would help you follow the plume an hour later?

Choose the time and the layer your map would need.

Raikoke, Kuril Islands, 22 June 2019. International Space Station photograph, NASA Johnson Space Center.

Could a model spot the smoke?

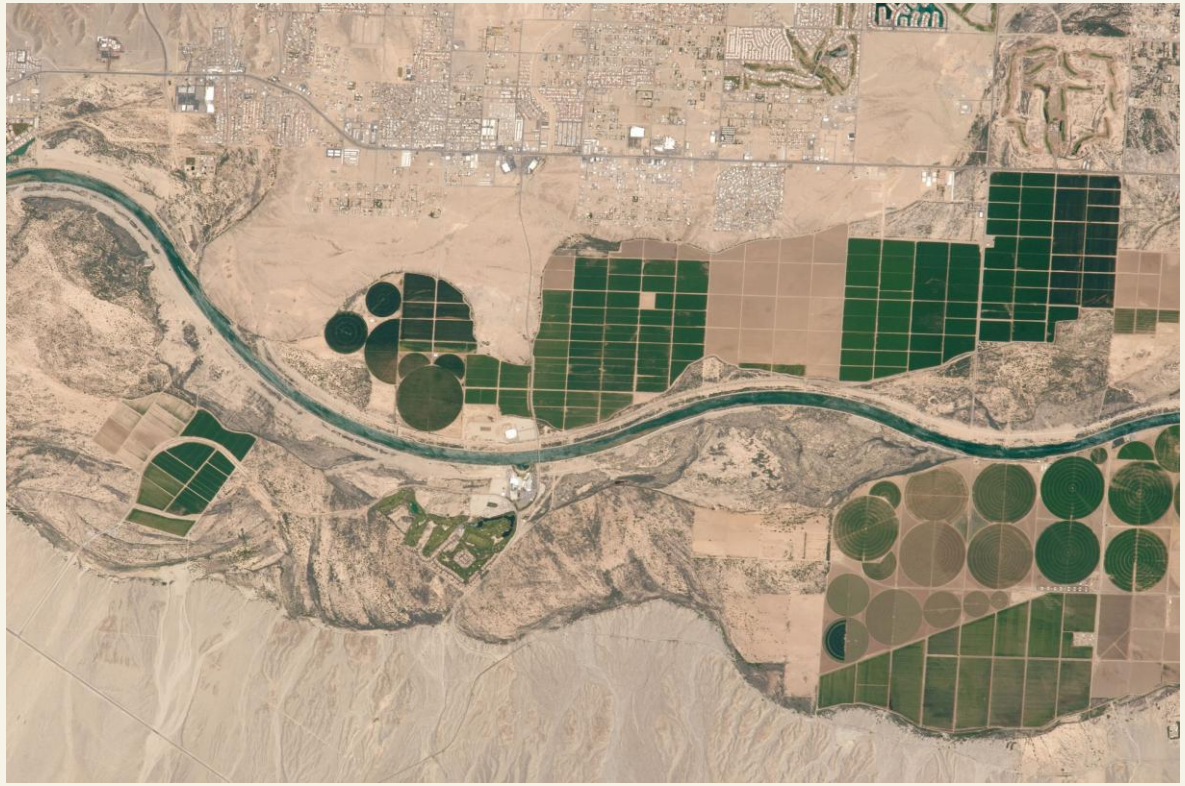
GeoAI uses models to recognize patterns in geographic data. Could one outline the smoke in this image?



How would you catch a cloud marked as smoke?

Amazon fires, 19 August 2019 • NASA Earth Observatory / Joshua Stevens, Terra MODIS.

What would you map from home?



Choose a place someone at your table knows. Tell a partner who has a decision to make there.

"I'd compare ____ with ____ to find out ____."

Name one observation that could change your answer.

Fields along the Colorado River, Fort Mojave Indian Reservation, 14 April 2017. International Space Station photograph, NASA Johnson Space Center.

Image and map sources

NASA's original captions, credits and downloads

Earth and the iceberg

Apollo 8 · Earthrise

San Francisco region at night

A-23A · blue meltwater from two views

A-23A · seafloor and satellite views

A-23A · track map and final fragments

Other landscapes

Camp Fire, California

Raikoke volcano, Kuril Islands

Colorado River agriculture

Workshop tools

NASA EONET v3 data guide

Esri ArcGIS Maps SDK for JavaScript

Image dates on the slides refer to the observations. The hands-on exercise uses the saved EONET sample from 12 November 2026.

Sources for the new investigations

Earth from above

Willamette River elevation map · Oregon DOGAMI

What is a DEM? · USGS

Amazon fires near Novo Progresso · NASA

Space Shuttle photograph of UARS · NASA

Pacific atmospheric river · NASA SVS

Build and extend the map

A-23A event record · NASA EONET

ArcGIS Pro · Esri

Map graphics in JavaScript · Esri

iOS samples · Esri Swift

Android samples · Esri Kotlin

Nearby Search · Google Maps Platform

Find observation dates in the image captions. Open the source links for instrument details and credits.

GIS and mobile field notes

GIS concepts

GeoJSON specification

Coordinate systems and projections

PostGIS spatial queries

Natural Earth map data

GeoAI examples

Mobile projects

My Esri · Get a native license

ArcGIS Maps SDK for Swift · 200.x

ArcGIS Maps SDK for Kotlin · 200.x

Google Places · Nearby Search

Expo SDK documentation

In the workshop folder, application contains the web map and mobile contains the three phone projects.

How could this work on a phone?

A field map

Switch between two dates on an Esri map in the Swift or Kotlin starter.

Swift · iOS / Kotlin · Android

A nearby-place search

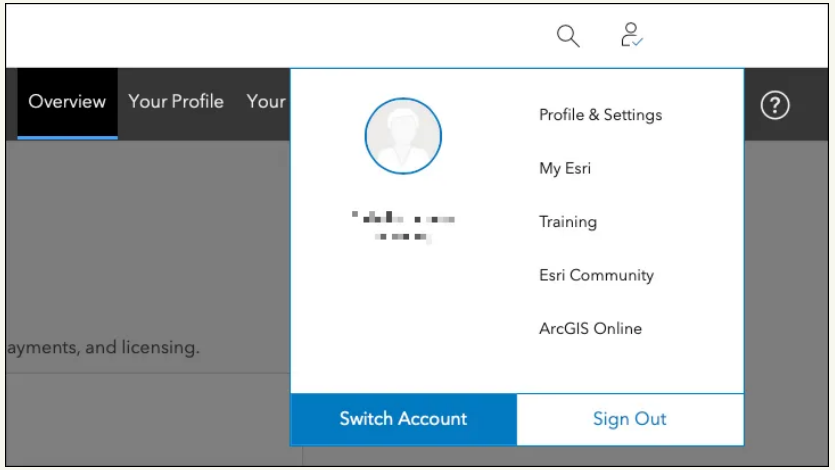
Choose a place type and a search radius around a point.

Google Places · Nearby Search

Choose a question to try on a phone. Which interaction would help a person in the field?

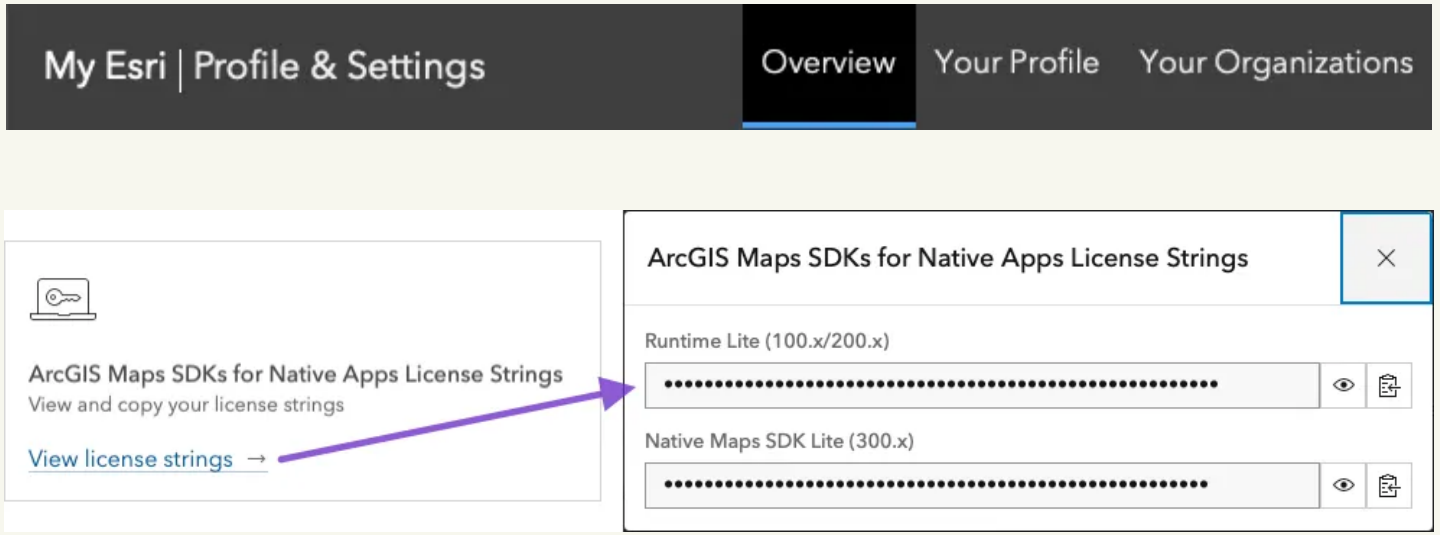
Find your Esri Lite license

Open the account menu



Choose Profile & Settings.

In Overview, open the license card



Choose View license strings, then copy the upper 100.x/200.x value for these projects.

[Open My Esri]

[Screenshots from Esri's native SDK licensing guide. The strings shown are masked.]

iOS / Move the iceberg one week

Open the Swift project

In Xcode, open IcebergMap.xcodeproj. Select the IcebergMap scheme and an iPhone simulator, then press Run.

Try both dates

Tap Show 6 March. The earlier position is farther west. Tap Show 13 March to return.



Starter and Mac test steps in mobile/ios-iceberg/README.md.

Esri Swift sample • reference

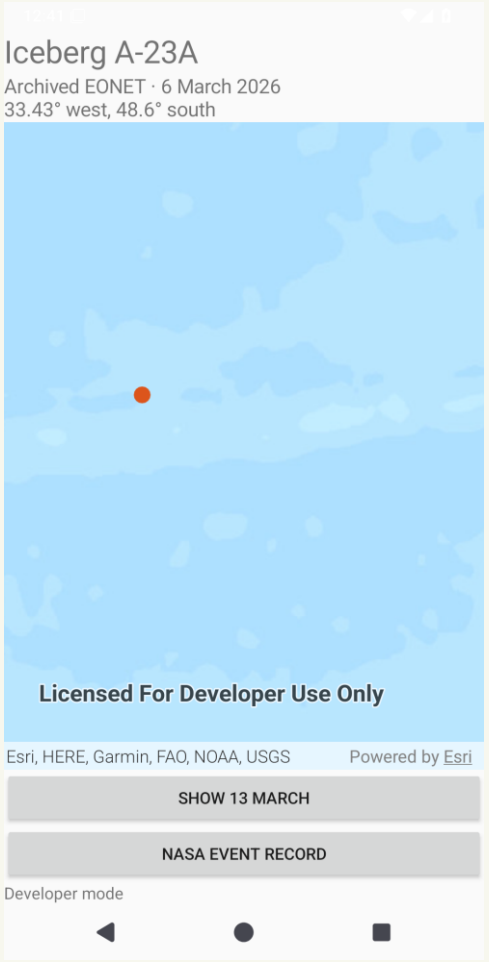
Android / Test the same prediction

Run Iceberg Map

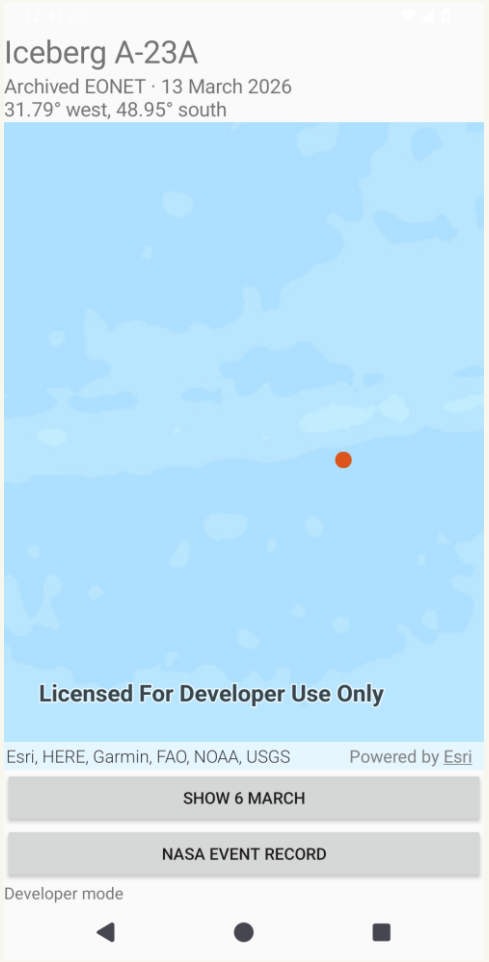
Open mobile/android-iceberg in Android Studio. Sync Gradle and run the app on an emulator or device.

Which way did it move?

Switch dates, then compare the orange markers. Moving east makes a western longitude less negative.



6 March



13 March

Both captures are from the workshop app on Android 15.

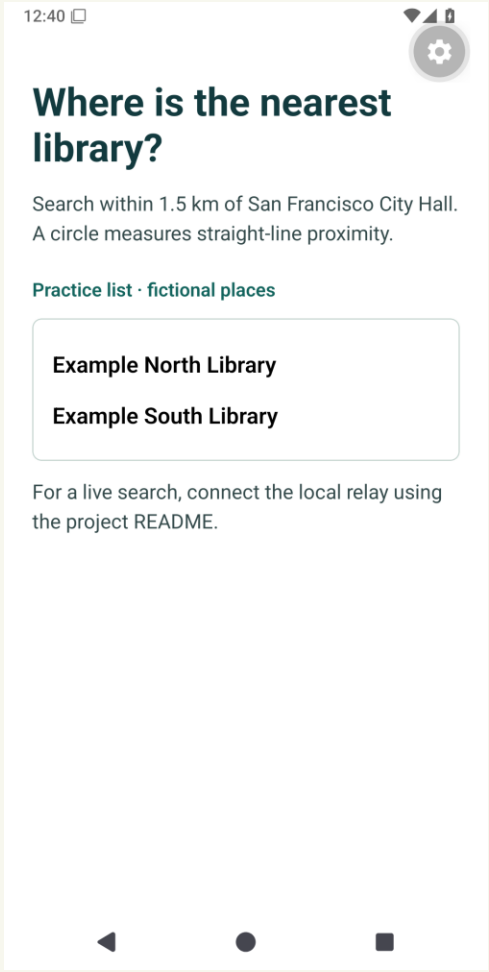
React Native / What is nearby?

Look for a library

The starter asks for libraries within 1.5 km of San Francisco City Hall. Run npm ci, then npm start in mobile/places-expo.

Read the result label

The two practice names are fictional. Connect the optional relay to run a live Google Places search.



The practice screen runs without a Google key.

Workshop app • Expo Go on Android

Thank you.

Questions?

What would you map first, and who would use the answer?